

Multi-Model Neural CDE Interpretability + Physics Audit

Full Interpretation + Derivation Reference (3-Way: Epoch 3 vs Epoch 4 vs Epoch 5)

Artifacts used: 3_model_comparison.json, 3_model_comparison.md, plots.zip (31 figures)

Models: epoch_3_013026, epoch_4_013026, epoch_5_013026

Samples: 3000 **Seed:** 42 **Input dim:** 54 features **Output heads:** 10

0. Notation and Conventions

Let the dataset be a multivariate time series with feature dimension $d = 54$. For each sample i , we take a window (sequence) of length T (the tool's internal slicing contract), forming a continuous-time path $X^{(i)}: [0,1] \rightarrow \mathbb{R}^d$ via interpolation.

We compare three checkpoints $M^{(3)}, M^{(4)}, M^{(5)}$ (Epoch 3/4/5). Each model produces a vector of head outputs:

$$\hat{\mathbf{y}} = (\hat{\mathbf{y}}_{\text{call_offset}}, \hat{\mathbf{y}}_{\text{put_offset}}, \hat{\mathbf{y}}_{\text{wing_width}}, \hat{\mathbf{y}}_{\text{dte}}, \hat{\mathbf{y}}_{\text{pop}}, \hat{\mathbf{y}}_{\text{roi}}, \hat{\mathbf{y}}_{\text{max_loss}}, \hat{\mathbf{y}}_{\text{confidence}}, \hat{\mathbf{y}}_{\text{entry}}, \hat{\mathbf{y}}_{\text{exit}}) \in \mathbb{R}^{10}.$$

Definitions

- d : number of input features (54).
 - T : sequence/window length.
 - X : continuous path representation of the windowed features.
 - $\hat{\mathbf{y}}$: model outputs across 10 heads.
 - $M^{(k)}$: model checkpoint for epoch $k \in \{3,4,5\}$.
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1. Data Contract, Leakage Masking, Scaling, and Why This Matters Physically

1.1 Data Contract

The audit assumes a strict I/O contract: feature order, head order, scaling method, and leakage masking are identical across checkpoints. Any silent drift in those contracts will produce “differences” that are not learning-based but pipeline-based.

1.2 Leakage Masking (Causality Constraint)

Leakage masking enforces that no future-dependent features leak information forward into the window. In physics terms: **we enforce a causal arrow of time** (no acausal coupling).

Formally, if a feature component $X_j(t)$ is known to be future-derived, we replace it by a masked version $\tilde{X}_j(t)$ such that $\tilde{X}_j(t)$ is measurable w.r.t. the sigma algebra $\mathcal{F}_t$ generated by data up to time t .

$$\tilde{X}_j(t) \in \mathcal{F}_t \text{ for all } t.$$

Definitions

- $X_j(t)$: feature j at time t .
- $\tilde{X}_j(t)$: masked feature (causal version).
- $\mathcal{F}_t$: information available up to time t .

1.3 Robust Scaling (Stability Under Heavy Tails)

Financial features are heavy-tailed; robust scaling reduces the influence of extreme events (jumps) on interpretability metrics.

A canonical robust transform is:

$$X'_j(t) = \frac{X_j(t) - \text{median}(X_j)}{\text{IQR}(X_j)}.$$

Definitions

- $X'_j(t)$: scaled feature.
- $\text{median}(X_j)$: median across calibration sample.
- $\text{IQR}(X_j) = Q_{0.75} - Q_{0.25}$: interquartile range.

2. Neural CDE Foundations (The Physics: Continuous-Time Dynamics)

2.1 Core Neural CDE Equation

A Neural CDE models latent state evolution driven by the control path $X(t)$:

$$Z(t) = Z(0) + \int_0^t f(Z(s); \theta) dX(s).$$

Definitions

- $Z(t) \in \mathbb{R}^h$: latent state (dimension h).
- $X(t) \in \mathbb{R}^d$: control path (input features).
- $f: \mathbb{R}^h \rightarrow \mathbb{R}^{h \times d}$: learned vector field.
- θ : parameters of the vector field network.

2.2 Discretized Approximation (Audit-Relevant)

When implemented numerically on a window $t_0 < \dots < t_T$, a common approximation resembles:

$$Z_{t_{n+1}} \approx Z_{t_n} + f(Z_{t_n}; \theta) (X_{t_{n+1}} - X_{t_n}).$$

Definitions

- Z_{t_n} : latent at discrete index n .
- $X_{t_{n+1}} - X_{t_n}$: observed increments (financial “impulses”).

2.3 Physical Interpretation

- $X(t)$ is the **driving signal** (market micro-dynamics).
- $f(Z; \theta)$ is a **state-dependent conductivity tensor** mapping input increments into latent motion.
- Stability/robustness questions become: how sensitive is $\hat{y}$ to perturbations in X , and how consistent are internal explanations across checkpoints?

3. What the Tool Computes (Mathematics + “Physics” Metrics)

You requested a precision reference spec—so below are the full derivations and interpretations for each module, including additional high-value metrics to make “best model selection” scientific rather than vibes-based.

3A. Permutation Importance (Global and Per-Head)

3A.1 Definition (General)

Given a metric $S(\cdot)$ that measures prediction quality (or a proxy score), permutation importance for feature j is:

$$PI_j = S(\cdot)(\hat{y}(X)) - \mathbb{E}_{\pi}[S(\cdot)(\hat{y}(X^{(\pi j)}))],$$

where $X^{(\pi j)}$ is X with feature j permuted across samples (destroying its association with the target while preserving its marginal distribution).

Definitions

- $S(\cdot)$: score function (audit-defined).
- $X^{(\pi j)}$: dataset with feature j permuted.
- π : random permutation operator across samples.
- PI_j : importance of feature j .

3A.2 Normalization to a Distribution (for Divergence Metrics)

To compare feature importance as a “probability-like” distribution:

$$p_j = \frac{\max(PI_j, 0)}{\sum_{m=1}^d \max(PI_m, 0)}.$$

Definitions

- p_j : normalized nonnegative importance mass for feature j .

3A.3 Per-Head Permutation Importance

For head h , compute $PI_j^{(h)}$ by scoring only that head:

$$PI_j^{(h)} = S_h(\cdot)(\hat{y}_h(X)) - \mathbb{E}_{\pi}[S_h(\cdot)(\hat{y}_h(X^{(\pi j)}))].$$

Definitions

- h : one of 10 output heads.
 - $S_h(\cdot)$: head-specific scoring function.
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3B. Surrogate Decision Trees (Rule Extraction for ROI Head)

We fit an interpretable model $g(\cdot)$ (a decision tree) to mimic the ROI head:

$$\hat{y}_{\text{roi}} \approx g(X_{\text{summary}}),$$

where X_{summary} is the flattened or aggregated representation used by the surrogate fitting routine.

3B.1 Fidelity (Explained Variance)

$$R^2 = 1 - \frac{\sum_{i=1}^n (\hat{y}_{\text{roi}}^{(i)} - g^{(i)})^2}{\sum_{i=1}^n (\hat{y}_{\text{roi}}^{(i)} - \bar{\hat{y}}_{\text{roi}})^2}.$$

Definitions

- n : number of audit samples.
- $\hat{y}_{\text{roi}}^{(i)}$: ROI prediction for sample i .
- $g^{(i)}$: surrogate prediction.
- $\bar{\hat{y}}_{\text{roi}}$: mean ROI prediction.

Interpretation

- Higher R^2 means the model's ROI decision surface is closer to a piecewise-constant rule system.
 - However, **too high** R^2 can also indicate **functional collapse** (model behaves like a shallow heuristic), which can be dangerous in nonstationary markets.
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3C. Divergence Metrics Between Models (Feature Importance Delta)

Let p and q be normalized importance distributions for two models.

3C.1 Cosine Similarity (Directional Alignment)

$$\text{CosSim}(p, q) = \frac{\langle p, q \rangle}{\|p\|_2 \|q\|_2}.$$

Definitions

- $\langle p, q \rangle = \sum_j p_j q_j$.
- $\|p\|_2 = \sqrt{\sum_j p_j^2}$.

3C.2 Jensen–Shannon Divergence (Information Distance)

$$\begin{aligned} \text{JSD}(p \parallel q) &= \frac{1}{2} \text{KL}(p \parallel m) + \frac{1}{2} \text{KL}(q \parallel m), m = \frac{p + q}{2}, \\ \text{KL}(p \parallel m) &= \sum_{j=1}^d p_j \log \left(\frac{p_j}{m_j} \right). \end{aligned}$$

Definitions

- m : mixture distribution.
- KL: Kullback–Leibler divergence.

3C.3 Wasserstein Distance (Earth Mover on Ranked Mass)

In 1D, Wasserstein is the L^1 distance between CDFs. For feature importance, the tool uses a practical proxy (rank-aligned or vectorized transport). The conceptual target:

$$W_1(p, q) = \inf_{\gamma \in \Gamma(p, q)} \int \|x - y\| d\gamma(x, y).$$

Definitions

- γ : transport plan.
- $\Gamma(p, q)$: couplings with marginals p, q .

3C.4 Rank Agreement (Spearman)

Let $r_p(j)$ and $r_q(j)$ be ranks of feature j . Then:

$$\rho_s = 1 - \frac{6 \sum_{j=1}^d (r_p(j) - r_q(j))^2}{d(d^2 - 1)}.$$

Definitions

- ρ_s : Spearman correlation of ranks.

3C.5 Top-K Overlap (Interpretability Intersection)

$$\text{Overlap@K}(A, B) = \frac{|A_K \cap B_K|}{K},$$

Definitions

- A_K : top K features for model A.
 - B_K : top K features for model B.
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3D. Physics-Inspired Output Stability Metrics (Per Head)

For a head h , compare two models' predictions on the same audit sample set: $u_i = \hat{y}_h^{(i)}(M^a)$, $v_i = \hat{y}_h^{(i)}(M^b)$.

3D.1 Mean Squared Difference (“Energy” / Response Work)

$$\text{MSD}_h = \frac{1}{n} \sum_{i=1}^n (u_i - v_i)^2.$$

Definitions

- MSD_h : mean squared deviation for head h .
- Physical analogy: squared displacement energy between response fields.

3D.2 Correlation (“Coherence” / Phase Locking)

$$r_h = \frac{\sum_{i=1}^n (u_i - \bar{u})(v_i - \bar{v})}{\sqrt{\sum_{i=1}^n (u_i - \bar{u})^2} \sqrt{\sum_{i=1}^n (v_i - \bar{v})^2}}.$$

Definitions

- r_h : Pearson correlation.
- Coherence interpretation: do the models move together sample-by-sample?

3D.3 Variance Ratio (“Amplitude Stability”)

$$\text{VR}_h = \frac{\text{Var}(u)}{\text{Var}(v)}.$$

Definitions

- $\text{VR}_h \approx 1$: similar output spread.
- $\text{VR}_h \ll 1$ or $\gg 1$: amplitude collapse or explosion.

3D.4 Kolmogorov–Smirnov Statistic (Distribution Shift)

$$D_h = \sup_x |F_u(x) - F_v(x)|.$$

Definitions

- F_u, F_v : empirical CDFs.
- D_h : maximum distributional discrepancy.

3D.5 Energy Distance (Distributional “Field Distance”)

$$\mathcal{E}(u, v) = 2\mathbb{E} \|U - V\| - \mathbb{E} \|U - U'\| - \mathbb{E} \|V - V'\|.$$

Definitions

- $U, U' \sim u, V, V' \sim v$ i.i.d.
- $\mathcal{E} = 0$ iff distributions match (under broad conditions).

3E. Gradient Saliency + Gradient Alignment (Sensitivity Geometry)

3E.1 Feature Saliency

For a head h , define saliency for feature j as an average gradient magnitude:

$$S_j^{(h)} = \mathbb{E}[|\frac{\partial \hat{y}_h}{\partial X_j}|].$$

Definitions

- $S_j^{(h)}$: sensitivity of head h to feature j .
- In physics terms: local response derivative (susceptibility).

3E.2 Temporal Saliency

Let $X(t)$ be discretized at $t_1, \dots, t_T$. Temporal saliency:

$$S_t^{(h)} = \mathbb{E}[\|\nabla_{X(t)} \hat{y}_h\|_1].$$

Definitions

- $S_t^{(h)}$: which time regions drive the decision most.

3E.3 Gradient Alignment Between Models

Let $g^a = \nabla_X \hat{y}_h(M^a)$, $g^b = \nabla_X \hat{y}_h(M^b)$. Then:

$$GA_h(M^a, M^b) = \frac{\langle g^a, g^b \rangle}{\|g^a\|_2 \|g^b\|_2}.$$

Definitions

- GA_h : gradient alignment cosine similarity.
- Interpretation: do models “look” at the same directions in feature space?

3F. SHAP Approximation (Ablation-Based Shapley Proxy)

Exact SHAP is combinatorial; the tool uses an approximation consistent with ablation logic.

A Shapley value for feature j conceptually is:

$$\phi_j = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} (f(S \cup \{j\}) - f(S)).$$

Definitions

- N : set of all features.
- S : subset coalition.
- $f(S)$: model output when only subset S is present (or others masked).

The report uses mean absolute SHAP mass:

$$\text{SHAP}_j = \mathbb{E}[|\phi_j|].$$

3G. Fisher Information (Parameter Sensitivity)

Let model likelihood be $p_\theta(y | x)$. Fisher information:

$$\mathcal{I}(\theta) = \mathbb{E}[\nabla_\theta \log p_\theta(Y | X) \nabla_\theta \log p_\theta(Y | X)^\top].$$

Practical layerwise proxy uses squared gradients of loss $\mathcal{L}$:

$$\mathcal{I}_\ell \approx \mathbb{E}[\|\nabla_{\theta_\ell} \mathcal{L}\|_2^2].$$

Definitions

- $\mathcal{I}_\ell$: Fisher proxy for layer/block ℓ .
- Physical analogy: local curvature/sensitivity (“stiffness”) of parameters.

3H. Mutual Information Between Features (Redundancy + Coupling)

For features X_i, X_j :

$$I(X_i; X_j) = \mathbb{E} \left[\log_{10} \frac{p(x_i, x_j)}{p(x_i)p(x_j)} \right].$$

Definitions

- $I(X_i; X_j)$: shared information (dependency).
 - Used to diagnose feature redundancy and coupling structure that can confound attribution.
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5. Model-by-Model Deep Interpretation (Pros/Cons + Physics Meaning)

Below uses the audit’s computed facts plus physically-grounded failure mode reasoning.

5.1 Epoch 3 (_{epoch_3_013026})

Attribution structure

- Top feature dominance:

$$\max_j p_j \approx 11.9\%.$$

- Entropy:

$$H(p) = - \sum_j p_j \log p_j \approx 3.30.$$

Interpretation

- Best feature balance of the three (most distributed reliance).
- Often implies better robustness under regime shifts (less single-factor fragility).

Surrogate fidelity

$$R^2 \approx 0.681.$$

Interpretation

- Strong but not extreme.
- Suggests ROI decision boundary is explainable but still retains nontrivial nonlinear structure.

Tree logic character

IVR drives the root, but secondary splits involve BB sigma dynamics and friction ratio—i.e., volatility geometry and microstructure.

Main risk

- SHAP vs permutation top-5 overlap is 4/5 (not perfect).
- Indicates some attribution instability in the top-5 boundary.

5.2 Epoch 4 (`epoch_4_013026`) ← Recommended Best Model

Attribution structure

- Top feature dominance:

$$\max_j p_j \approx 16.2\%.$$

- Entropy:

$$H(p) \approx 3.21.$$

Interpretation

- Slightly more concentrated than E3, but still not “single-feature collapse.”
- Concentration aligns with meaningful state variables (IVR/Greeks + risk pressure).

Surrogate fidelity

$$R^2 \approx 0.748.$$

Interpretation

- Higher fidelity than E3, meaning rules summarize ROI more reliably.

SHAP-Permutation consistency (critical)

Epoch 4 is the only model with:

$$\text{Top5}_{\text{perm}} = \text{Top5}_{\text{SHAP}}.$$

Interpretation

- This is the strongest “interpretability stability” signal in your audit.
- It means the model’s feature influence is not an artifact of the particular explanation method.

Fisher evolution meaning

The major parameter shift from E3→E4 is in regime detection and expert blocks—exactly where a multi-regime trading system should improve.

Epoch 4 reads like the model that learned the regime map cleanly without over-locking into IVR thresholds.

5.3 Epoch 5 (_{epoch_5_013026}) Why it is NOT the best

Surrogate fidelity is highest

$$R^2 \approx 0.801.$$

This looks good superficially, but the decision tree shows increasing self-referential IVR gating (multiple IVR thresholds), and attribution concentration increases:

- Top feature share:

$$\max_j p_j \approx 17.6\%.$$

- Entropy:

$$H(p) \approx 3.12(\text{lowest of the three}).$$

Interpretation

- Epoch 5 likely moved deeper into a basin where ROI is “explained” primarily by IVR and delta.
- This can be a *brittle heuristic* when IVR is noisy, delayed, or the regime is driven by flow/structure rather than implied vol regime.

Feature substitution warning

Epoch 5 top-5 swaps out `gap_risk_score` and replaces it with `pressure_up`.

Interpretation

- That is a qualitative strategy shift: from gap/jump risk awareness to a more directional pressure component.
- For iron condors, neglecting gap risk is one of the classic “looks great until it dies” failure modes.

6. Final Recommendation: Best Model and Scientific Justification

Best checkpoint to deploy (given the evidence in the attached artifacts)

Use `epoch_4_013026` as the primary candidate (with `epoch_3_013026` as a robustness fallback).

Why Epoch 4 is best (scientific criteria)

Epoch 4 optimizes the three most important “science-grade” properties of a deployable trading model:

1. **Interpretability consistency across methods**

$$\text{Top5}_{\text{perm}} = \text{Top5}_{\text{SHAP}}(\text{only Epoch 4}).$$

2. **High surrogate fidelity without obvious degeneracy**

$R^2(E4)$
= 0.748 is high enough for reliable rules, but not the extreme IVR self-gating seen in E5’s tree.

3. **Parameter sensitivity evolution where it matters**

Fisher indicates E3→E4 improved regime/expert logic substantially; E4→E5 changes are smaller and trend toward more rigid gating.

Why Epoch 5 is not best

- The tree becomes too IVR-threshold dominated (risk of single-factor fragility).
- Entropy falls and top-feature dominance rises.
- It deprioritizes `gap_risk_score` (a key survival variable for short-vol structures).

Why Epoch 3 is not best

- Lower surrogate fidelity and less perfect attribution agreement (4/5 top-5 overlap vs 5/5 in E4).
- Fisher indicates regime detector/expert routing is still materially under-trained relative to E4.

7. “Refine This to a Science” — Required Upgrades to Make the Audit Conclusive

To make “best model selection” *scientific*, the audit must eliminate the major uncertainty modes.

7.1 Enforce Sample-Index Identity Across All Modules

Persist a single index set $I = \{i_1, \dots, i_n\}$ and reuse it for:

- permutation importance,
- stability,
- gradients,

- SHAP,
- Fisher.

Then coherence/correlation becomes physically meaningful:

r_h is only interpretable if (u_i, v_i) correspond to the same stimulus $X^{(i)}$.

7.2 Replace Coarse Ranking With Continuous Scoring

Define continuous robustness/fidelity/balance scores:

(a) Balance score (entropy and dominance)

$$\text{Balance}(M) = \lambda_1 \frac{H(p)}{\log d} - \lambda_2 \max_j p_j.$$

(b) Interpretability score

$$\text{Interp}(M) = R_{\text{tree}}^2(M) - \eta \cdot \text{DepthPenalty}(M).$$

(c) Attribution consistency score

$$\text{Consist}(M) = \frac{|\text{TopK}_{\text{perm}} \cap \text{TopK}_{\text{SHAP}}|}{K}.$$

(d) Stability under perturbations (true physics test)

For small perturbations δX (noise, regime shifts), define:

$$\text{Lip}_h(M) \approx \mathbb{E} \left[\frac{\| \hat{y}_h(X + \delta X) - \hat{y}_h(X) \|}{\| \delta X \|} \right].$$

A robust model has lower Lip_h for risk heads (ROI, MAX_LOSS) under realistic perturbations.

8. Additional Metrics to Add (High Value, Low Bullshit)

To go beyond the current tool (and remove ambiguity from stability plots), add:

8.1 Distance Correlation (Nonlinear Dependence)

$$\text{dCor}(U, V) = \frac{\text{dCov}(U, V)}{\sqrt{\text{dVar}(U)\text{dVar}(V)}}.$$

Use this instead of Pearson when outputs are nonlinear/clipped.

8.2 Maximum Mean Discrepancy (MMD) for Output Distributions

$$\text{MMD}^2(P, Q) = \mathbb{E}[k(U, U')] + \mathbb{E}[k(V, V')] - 2\mathbb{E}[k(U, V)].$$

This is a strong two-sample test and behaves well in high dimensions.

8.3 Regime-Conditional Stability

Compute stability metrics conditional on regime clusters $\mathcal{C}$ (from your regime detector):

$$\text{MSD}_h^{(c)} = \frac{1}{|\mathcal{C}_c|} \sum_{i \in \mathcal{C}_c} (u_i - v_i)^2.$$

This answers: “are they different only in chaos regimes?”

8.4 Calibration Against Realized Outcomes (When Available)

Ultimately, for trading, interpretability must connect to realized performance:

$$\text{CalErr} = \mathbb{E}[(\hat{y}_{\text{roi}} - \text{ROI}_{\text{real}})^2] \text{ and } \text{Brier} = \mathbb{E}[(\hat{y}_{\text{pop}} - \mathbf{1}_{\text{profit}})^2].$$

9. Final Deliverable Summary (What We Now Have)

- A mathematically rigorous interpretation of:
 - permutation + per-head attributions,
 - surrogate trees + fidelity theory,
 - divergence metrics (CosSim, Spearman, JSD, Wasserstein, overlap),
 - physics-inspired stability (MSD, coherence, amplitude, KS, energy distance),
 - gradients + alignment,
 - SHAP proxy,
 - Fisher sensitivity evolution,
 - MI redundancy structure,
 - plus scientifically necessary additions to remove ambiguity.

- A clear model selection:
Epoch 4 is best because it is the only checkpoint with perfect attribution method agreement at the top, high surrogate fidelity without the increased IVR threshold dominance seen in Epoch 5, and parameter sensitivity evolution concentrated in regime/expert modules where learning matters most.

Appendix A — Feature Registry (d = 54)

Feature Dictionary (54)

Feature	Meaning	Canonical definition / equation (or algorithm)	Notes / interpretation
open	Spot open price at bar t	O_t	Base OHLCV.
high	Spot high price at bar t	H_t	Base OHLCV.
low	Spot low price at bar t	L_t	Base OHLCV.
close	Spot close price at bar t	C_t	Base OHLCV.
volume	Spot traded volume at bar t	V_t	Often log-scaled; impacts liquidity/flow proxies.
δ	Option delta (directional sensitivity)	$\Delta = \frac{\partial P}{\partial S}$	Usually aggregated across chosen strikes/legs; sign depends on calls/puts.
γ	Option gamma (curvature sensitivity)	$\Gamma = \frac{\partial^2 P}{\partial S^2}$	High Γ means fast delta changes; risk spikes near ATM.
ν	Option vega (vol sensitivity)	$\nu = \frac{\partial P}{\partial \sigma}$	Exposure to implied vol changes; iron condors are typically short vega.
θ	Option theta (time decay)	$\Theta = \frac{\partial P}{\partial t}$	For short premium, theta

			is typically “positive carry” (benefits from decay).
iv	Implied volatility level	σ_{impl} such that $BS(S, K, r, q, \sigma, T) = P_{\text{mkt}}$	Solved by numeric root-finding (Newton/bisection).
ivr	Implied volatility rank (0–1)	IVR $= \frac{\text{IV} - \min(\text{IV}_{\text{lookback}})}{\max(\text{IV}_{\text{lookback}}) - \min(\text{IV}_{\text{lookback}})}$	Normalizes IV into historical range; key state for premium strategies.
spread_ratio	Bid–ask microstructure cost proxy	$\text{spread_ratio} = \frac{\text{ask} - \text{bid}}{\text{mid} + \varepsilon}, \text{mid} = \frac{\text{ask} + \text{bid}}{2}$	High spread_ratio = worse fills / slippage regime.
te	“Time exposure” / time-to-expiry measure used internally	Common: $\text{TE} = \frac{\text{DTE}}{365}$ or $\sqrt{\frac{\text{DTE}}{365}}$	If your code defines TE differently, it must match training. Often used to scale Greeks.
strike	Strike level proxy (selected strike or aggregate)	K	Often normalized by spot: K/Spot or moneyness.
target_spot	Leakage column (future-dependent target)	(Should be masked)	Must be zeroed in audit and training-time contract.
max_dd_60m	Leakage column (future drawdown proxy)	(Should be masked)	Must be zeroed; otherwise leakage.
log_return	Log return of spot	$r_t = \ln\left(\frac{C_t}{C_{t-1}}\right)$	Base stationary-ish return.
vol_ewma	EWMA volatility estimate	$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_t^2$	Typical $\lambda \in [0.94, 0.99]$; may use sqrt at end.

ret_z	Z-score of return	$\text{ret_}z_t = \frac{r_t - \mu_w}{\sigma_w + \varepsilon}$	μ_w, σ_w computed over rolling window w .
atr_pct	ATR as percent of price	$\text{ATR}_t = \text{RMA}_n(\text{TR}_t),$ (\text{TR}\}_t = \max(H_t - L_t,	$H_t - C_{t-1}$
kappa_proxy	“Kappa” proxy (tail/curvature/vol-of-vol proxy)	Common: rolling kurtosis or tail ratio; e.g. $\kappa = \frac{E[(r-\mu)^4]}{(E[(r-\mu)^2])^2}$	Name suggests kurtosis-like tail heaviness; depends on your implementation.
vol_energy	Volatility “energy” proxy	Common: r_t^2 or rolling mean of r^2 : $\text{vol_energy} = \text{MA}_n(r^2)$	Physically: instantaneous kinetic energy of price path.
rsi_dyn	Dynamic RSI (adaptive length or smoothing)	Standard RSI: $RS = \frac{\text{RMA}(\text{gain}, n)}{\text{RMA}(\text{loss}, n)},$ $RSI = 100 - \frac{100}{1 + RS}$	“dyn” implies adaptive <i>nor</i> adaptive RMA factor.
adx_adaptive	Adaptive ADX (trend strength)	Compute $+DI, -DI$ then $(DX = 100 \frac{\{$	$+DI - (-DI)$
psar_adaptive	Adaptive Parabolic SAR	PSAR recursion with AF adaptive: $PSAR_t = PSAR_{t-1} + AF(EP - PSAR_{t-1})$	Adaptive AF / step sizing to regime.
bb_mu_dyn	Dynamic Bollinger mean	$\mu_t = \text{MA}_n(C_t)$	“dyn” means <i>n</i> adapts.
bb_sigma_dyn	Dynamic Bollinger std	$\sigma_t = \sqrt{\text{MA}_n((C - \mu)^2)}$	Used for band width and regime.
bb_lower_dyn	Dynamic lower band	$BB_t^- = \mu_t - k\sigma_t$	Usually $k = 2$.
bb_upper_dyn	Dynamic upper band	$BB_t^+ = \mu_t + k\sigma_t$	Usually $k = 2$.
stoch_k_dyn	Dynamic stochastic %K	$\%K_t = 100 \frac{C_t - \min(L_{t-n..t})}{\max(H_{t-n..t}) - \min(L_{t-n..t}) +$	“dyn” implies adaptive lookback n .
consolidation_score	Consolidation / range-bound score	Common: inverse of bandwidth / ADX; e.g. $\text{consol} = \text{sigmoid}(a - \text{bw})$ or $\propto \frac{1}{\sigma}$	Implementation-specific; typically high when price compresses.
breakout_score	Breakout / expansion score	Common: bandwidth expansion + range break; e.g. $\text{breakout} \propto \Delta \text{BW} + \mathbf{1}[C_t > BB^+]$	Implementation-specific; high during expansion / regime change.

bb_percentile	Bollinger percentile position	$bb_pct = \frac{C_t - BB_t^-}{(BB_t^+ - BB_t^-) + \varepsilon}$	0 near lower band, 1 near upper band.
bw_expansion_rate	Bandwidth expansion rate	$BW_t = \frac{BB_t^+ - BB_t^-}{\mu_t + \varepsilon}; bw_exp = BW_t - BW_{t-1} \text{ (or \% change)}$	Detects volatility regime change.
cmf	Chaikin Money Flow	$MFV_t = \frac{(C_t - L_t) - (H_t - C_t)}{(H_t - L_t) + \varepsilon} V_t; CMF = \frac{\sum MFV}{\sum V} \text{ over window}$	Flow / accumulation proxy.
pressure_up	Upward “pressure” proxy	Often built from order-flow + price position; example: $pressure_up = EMA(\max(r_t, 0), n)$	Your pipeline likely uses engineered pressure from multiple signals.
pressure_down	Downward “pressure” proxy	Example: $pressure_down = EMA(\max(-r_t, 0), n)$	Often paired with <code>pressure_up</code> for asymmetry.
friction_ratio	Friction / slippage / noise ratio	Common: $\text{\texttt{friction}} = \frac{\text{\texttt{realized_range}}}{\text{\texttt{net_move}}}$	<code>\text{\texttt{net_move}}</code>
exec_allow	Execution allow flag/gate	$exec_allow \in \{0,1\}$ from rule engine	If 0, execution should be blocked; used as a hard constraint feature.
gap_risk_score	Gap / jump risk score	Standard gap: $G = \frac{O_t - C_{t-1}}{\dots}$	$O_t - C_{t-1}$
risk_override	Manual/system risk override	$risk_override \in \{0,1\}$	Safety switch / supervisory input.
iv_confidence	Confidence in IV estimate / IV surface quality	Typically derived from spread, quote depth, fit error, or sample count; e.g. $iv_conf = \exp(-c \cdot fit_err)$	Higher = better IV measurement reliability.
mtf_consensus	Multi-timeframe consensus indicator	$mtf = \text{agg}(\text{signals}_{1m}, \text{signals}_{5m}, \text{signals}_{15m})$	Usually a weighted average or vote of regime states.
macd_norm	Normalized MACD line	$MACD = EMA_{fast}(C) - EMA_{slow}(C)$; normalize: $\frac{MACD}{\sigma(C) + \varepsilon}$ or by ATR	Normalization makes scale comparable across regimes.
macd_signal_norm	Normalized MACD signal	$Signal = EMA(MACD, n)$; normalized similarly	Used with histogram to

			detect acceleration.
macd_histogram	MACD histogram	$Hist = MACD - Signal$	Momentum acceleration proxy.
plus_di	+DI (Directional Indicator)	$+DM_t = \max(H_t - H_{t-1}, 0)$ if greater than (	$L_t - L_{t-1}$
minus_di	-DI (Directional Indicator)	$-DM_t = \max(L_{t-1} - L_t, 0)$ if greater than (	$H_t - H_{t-1}$
psar_trend	PSAR trend state	Often binary: 1 uptrend, 0 downtrend (or ± 1) from PSAR logic	Encodes regime state from SAR.
psar_reversion_mu	PSAR reversion mean level	Implementation-specific; often rolling mean of (price - PSAR) or mean reversion anchor	Used to quantify distance to SAR anchor / reversion pressure.
beta1_norm_stub	Normalized beta proxy / placeholder	If “stub”: constant or low-signal; otherwise $\beta = \frac{\text{Cov}(r_{asset}, r_{bench})}{\text{Var}(r_{bench})}$ normalized	If it’s a stub, it should be near 0 importance.
chaos_membership	Chaos/regime membership (fuzzy)	Often membership function $\mu_{\text{chaos}}(x) \in [0,1]$ based on volatility spikes / entropy / fractal metrics	High means unstable regime; should reduce condor activity.
position_size_mult	Position size multiplier	$\text{size_mult} = f(\text{risk}, \text{confidence}, \text{regime})$	Used to scale exposure, typically $[0, 1]$ or wider.
fuzzy_reversion_11	Fuzzy mean-reversion membership	Membership function $\mu_{\text{revert}}(x) \in [0,1]$ using RSI/BB/ADX etc.	“11” suggests a specific rule index/version.

A few important notes (so the table is *actually* useful for engineering)

1. Greek features ($\delta, \gamma, \nu, \theta$)

If these are computed from a portfolio of legs (iron condor), they should be **portfolio Greeks**:

$$\Delta_{\text{port}} = \sum_{\ell} w_{\ell} \Delta_{\ell}, \Gamma_{\text{port}} = \sum_{\ell} w_{\ell} \Gamma_{\ell}, \nu_{\text{port}} = \sum_{\ell} w_{\ell} \nu_{\ell}, \Theta_{\text{port}} = \sum_{\ell} w_{\ell} \Theta_{\ell}$$

- w_ℓ = signed quantity (+ long, – short) for leg ℓ .
- 2. **“Dynamic/adaptive” indicators**
Anything with `_dyn` or `_adaptive` has parameters that are *not universal*. The equations above are standard; your implementation must lock down:
 - adaptive window selection rule,
 - smoothing constants,
 - normalization scheme.
- 3. **Score features (`consolidation_score`, `breakout_score`, `pressure_up/down`, `friction_ratio`, `gap_risk_score`)**
These are often **bespoke composite features**. If you want, paste the exact functions from your feature builder and I’ll rewrite the table entries so the formulas match your code 1:1.

Feature Physics Taxonomy + Expected Iron-Condor Effect

Goal: for each feature, classify (i) the *physical state variable* it represents in the market+options system, and (ii) the *directional expectation* for an iron condor (short premium) decision: **GOOD** / **BAD** / **CONTEXT** (and why).

Legend (Effect on short-premium condor viability)

- **GOOD** = tends to improve expected condor outcomes (higher POP, better decay capture, lower tail risk)
- **BAD** = tends to worsen outcomes (gap risk, trend breakout risk, unstable microstructure, adverse skew dynamics)
- **CONTEXT** = sign depends on regime / distance-to-spot / hedging geometry; must be interpreted with other variables

Category Key (Physics labels)

- **P** = Price state (spot kinematics)
 - **RV** = Realized volatility / dispersion (path energy)
 - **IV** = Implied vol surface state (risk-neutral volatility field)
 - **GR** = Greeks / sensitivity geometry (surface derivatives)
 - **MF** = Market flow / microstructure friction (execution + order-flow)
 - **RG** = Regime / phase indicator (state classification, chaos, multi-timeframe)
 - **EX** = Execution constraints / safety gates (hard controls)
 - **RR** = Risk / tail-jump metrics (gap, drawdown proxies, overrides)
-

Table A — Physics Category + Expected Condor Effect

Feature	Physics category	What it “is” physically	Expected condor effect	Why (mechanistic)
open	P	Position of spot at bar start	CONTEXT	Needed for state; not directionally “good/bad” alone.
high	P/RV	Intrabar max displacement	BAD (if elevated)	Larger intrabar excursions increase touch/breach risk of wings.
low	P/RV	Intrabar min displacement	BAD (if elevated range)	Same as high: larger range increases short-strike touch risk.
close	P	End-of-bar spot state	CONTEXT	Used for returns and levels; sign depends on structure.
volume	MF	Liquidity/participation proxy	CONTEXT	High volume can mean stable liquidity (good fills) or panic flow (bad).
δ	GR	First derivative wrt spot	CONTEXT	For condors you want near-delta-neutral; high net (
γ	GR	Curvature wrt spot	BAD (if high)	High gamma means delta changes fast; short structures become unstable near ATM.
ν	GR/IV	Sensitivity wrt IV	CONTEXT→often BAD if short vega	Condors are typically short vega; IV expansion hurts.
θ	GR/time	Sensitivity wrt time decay	GOOD (if favorable sign)	Short premium benefits from time decay, assuming range-bound regime.
iv	IV	Instant implied vol level	CONTEXT	High IV increases premium (good)

				but also tail risk (bad).
ivr	IV/RG	IV position within historical range	CONTEXT	High IVR = rich premium (good) but often crisis regime (bad). Needs filters.
spread_ratio	MF	Relative bid–ask friction	BAD (if high)	High spreads increase slippage; makes backtest→live drift worse.
te	IV/time	Time-to-expiry scaling field	CONTEXT	Short DTE increases gamma risk; longer DTE reduces theta/day; balance needed.
strike	P/GR	Geometric position of strikes	CONTEXT	Must be interpreted via moneyness and wing width; not “good/bad” alone.
target_spot	RR (leakage)	Future-derived (acausal)	MUST BE ZERO	Any nonzero use is leakage; destroys validity.
max_dd_60m	RR (leakage)	Future drawdown proxy	MUST BE ZERO	Leakage; creates false stability.
log_return	P/RV	Instant velocity in log space	BAD (if large magnitude)	Large returns imply instability; condor needs low drift over holding horizon.
vol_ewma	RV	Smoothed kinetic energy scale	BAD (if high)	High realized vol increases wing touch probability and vega/gamma stress.
ret_z	RV/RG	Standardized shock intensity	BAD (if large (	z
atr_pct	RV	Range scale normalized by price	BAD (if high)	ATR is realized “temperature”; condors prefer lower temperature.
kappa_proxy	RV/RR	Tail-heaviness / kurtosis proxy	BAD (if high)	High kurtosis = fat tails; condor tail loss risk rises.

vol_energy	RV	Squared-return energy density	BAD (if high)	Same physical meaning as temperature/energy; more motion → more breach risk.
rsi_dyn	P/RG	Oscillator (overbought/oversold)	CONTEXT	Overbought/oversold can favor reversion (good) or signal trend momentum (bad).
adx_adaptive	RG	Trend strength (phase coherence)	BAD (if high)	High ADX means trend regime; condors get run over by sustained drift.
psar_adaptive	RG	Trend-following phase state	BAD (if strong trend)	Trend-following indicator; strong signal implies directional regime.
bb_mu_dyn	P/RG	Mean field / equilibrium level	CONTEXT	Useful as reversion anchor; sign depends on displacement from mean.
bb_sigma_dyn	RV	Dispersion of price around mean	BAD (if high)	Wider bands mean higher vol; condors face more boundary hits.
bb_lower_dyn	P/RG	Lower boundary field	CONTEXT	Matters vs close; crossing can imply breakdown (bad) or oversold (revert good).
bb_upper_dyn	P/RG	Upper boundary field	CONTEXT	Same: breakout vs reversion depends on regime (ADX, pressure).
stoch_k_dyn	P/RG	Position in recent high-low range	CONTEXT	Extremes can mean reversion opportunity or breakout momentum.
consolidation_score	RG	“Low entropy / compressed” regime	GOOD (if high)	Consolidation implies bounded motion; ideal for short premium.

breakout_score	RG/RR	“Phase transition / expansion” signal	BAD (if high)	Breakout = regime change with drift; condors vulnerable.
bb_percentile	P/RG	Normalized position within bands	CONTEXT	Mid-range is best; near extremes can be reversion good or breakout bad.
bw_expansion_rate	RG/RV	Volatility expansion derivative	BAD (if high positive)	Rapid widening = phase transition to high-vol regime.
cmf	MF	Accumulation/distribution (flow)	CONTEXT	Strong directional flow can drive trend (bad), but stable flow can reduce noise (mixed).
pressure_up	MF/RG	Upward force proxy	BAD (if high) for short calls	Sustained up pressure threatens call wing; may be okay if balanced with down pressure.
pressure_down	MF/RG	Downward force proxy	BAD (if high) for short puts	Sustained down pressure threatens put wing; tail risk rises.
friction_ratio	MF/RG	Choppiness / path roughness	CONTEXT	High friction can mean mean-reverting chop (good) or noisy whipsaw (bad).
exec_allow	EX	Hard execution gate	GOOD (if 1)	If 0, you should not trade; prevents invalid deployments.
gap_risk_score	RR	Jump / discontinuity likelihood	BAD (if high)	Condors hate jumps; gap risk dominates tail losses.
risk_override	EX/RR	Supervisor override gate	CONTEXT	If 1, likely means “reduce/stop”; operationally protective.
iv_confidence	IV/MF	Reliability of IV estimate	GOOD (if high)	Low confidence means IV is noisy

				→ Greeks unreliable → bad hedging decisions.
mtf_consensus	RG	Multi-timeframe regime agreement	GOOD (if stable / range)	Consensus can reduce false positives; but if consensus is “trend”, condor is bad.
macd_norm	RG	Normalized momentum field	BAD (if strong trend)	Strong momentum implies drift regime; condor vulnerable.
macd_signal_norm	RG	Smoothed momentum	BAD (if strong)	Same as macd_norm; strengthens trend evidence.
macd_histogram	RG/RV	Momentum acceleration	BAD (if large magnitude)	Acceleration often precedes breakouts; not condor-friendly.
plus_di	RG	Positive directional movement	BAD (if dominates)	Indicates persistent upward directional regime.
minus_di	RG	Negative directional movement	BAD (if dominates)	Indicates persistent downward directional regime.
psar_trend	RG	Trend state flag	BAD (if trend)	Condors prefer neutral/sideways rather than trend phases.
psar_reversion_mu	P/RG	Reversion anchor around PSAR	CONTEXT	Useful if it truly predicts reversion; otherwise could lag.
beta1_norm_stub	RG (macro)	Market-beta coupling proxy	CONTEXT	High beta implies market-driven risk; could be neutral if hedged, bad if not.
chaos_membership	RG/RR	Chaotic regime membership	BAD (if high)	Chaos = instability / nonstationarity; condors have fat-tail exposure.

position_size_mu lt	EX/RR	Exposure scaling field	CONTEXT	High multiplier increases risk; should shrink under chaos/high gap risk/high trend.
fuzzy_reversion_11	RG	Mean-reversion membership	GOOD (if high)	Reversion regime supports premium harvesting if volatility not expanding.

Table B — “Expected Sign” of Each Feature on Key Condor Outcomes

Here are the three outcomes that actually matter mechanically:

1. **POP** (probability of profit)
2. **Tail loss risk** (max loss / breach probability)
3. **Execution realism** (slippage & fill quality)

I’ll denote expected qualitative direction as ↑ (increases), ↓ (decreases), or ↔ (context-dependent).

Feature	POP	Tail risk	Execution realism	Core reason
ivr	↔	↑ when high	↔	Rich premium but crisis regimes.
theta	↑	↔	↔	Higher decay helps if price stays bounded.
gamma	↓	↑	↔	High curvature amplifies boundary hits.
gap_risk_score	↓	↑↑	↔	Jump discontinuities dominate tail losses.
breakout_score	↓	↑	↔	Phase transition to drift/expansion.
consolidation_score	↑	↓	↔	Bounded motion regime.
bw_expansion_rate	↓	↑	↔	Vol expansion means more strikes touched.
atr_pct / vol_ewma / vol_energy	↓	↑	↔	Higher realized dispersion.
spread_ratio	↔	↔	↓	Direct slippage / worse fills.

iv_confidence	↔	↔	↑	Reliable IV → reliable Greeks → better execution decisions.
pressure_up/down	↓ when imbalanced	↑	↔	Persistent force threatens one wing.
adx_adaptive / di / macd	↓	↑	↔	Trend regime hurts condors.
chaos_membership	↓	↑↑	↔	Nonstationary risk; tail events more likely.
position_size_mult	↔	↑ if large	↔	Exposure scaling; must shrink in risk regimes.

Table C — Practical “Gating Rules” (Physics-informed)

These are not “signals” but **operational physics constraints** that tell you when a condor is structurally unsafe.

C1. Vol expansion gate

If $\text{fbw_expansion_rate} > \tau_{\text{bw}}$, then block or reduce size.

- τ_{bw} : expansion threshold.

C2. Trend gate

If $\text{adx_adaptive} > \tau_{\text{adx}}$, then condor is high-risk.

C3. Jump-risk gate

If $\text{gap_risk_score} > \tau_{\text{gap}}$, then avoid short premium.

C4. Microstructure gate (live realism)

If $\text{spread_ratio} > \tau_{\text{spr}}$, then expected slippage invalidates edge.

C5. Chaos gate

If $\text{chaos_membership} > \tau_{\text{chaos}}$, then shrink position_size_mult.

Definitions

- $\tau_{bw}, \tau_{adx}, \tau_{gap}, \tau_{spr}, \tau_{chaos}$: tuned thresholds calibrated from backtest-to-live drift analysis.

Table D — Feature Families (so the model can be audited by “state blocks”)

This is how a quant/risk reviewer will think about it:

Options surface / Greeks block

$\{\delta, \gamma, \nu, \theta, iv, ivr, strike, te, iv_confidence\}$

Price/return block

$\{open, high, low, close, log_return, ret_z\}$

Realized volatility / energy block

$\{atr_pct, vol_ewma, vol_energy, bb_sigma_dyn, bw_expansion_rate, \kappa_{proxy}\}$

Regime/trend/momentum block

$\{adx_adaptive, plus_di, minus_di, macd_norm, macd_signal_norm, macd_histogram, psar_adaptive, psar_trend, mtf_consensus, ch\}$

Mean-reversion / boundedness block

$\{bb_mu_dyn, bb_lower_dyn, bb_upper_dyn, bb_percentile, stoch_k_dyn, fuzzy_reversion_11, consolidation_score\}$

Flow + friction + execution block

$\{volume, spread_ratio, cmf, pressure_up, pressure_down, friction_ratio, exec_allow\}$

Tail risk / overrides block

$\{gap_risk_score, risk_override, position_size_mult\}$

Leakage block (must be masked)

$\{target_spot, max_dd_60m\}$